QUBO Formulation: Weighted Max-2-SAT

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Problem Statement. Given a set of n Boolean variables and a collection $\mathcal{C}$ of weighted clauses, where each clause contains at most two literals, the objective is to assign truth values to all variables so as to maximize the total weight of satisfied clauses. A clause is satisfied if at least one of its literals evaluates to true under the assignment.

Decision Variables. Let $x_i \in \{0, 1\}$ for $i \in \{1, \ldots, n\}$ denote the binary decision variables, where $x_i = 1$ represents the variable being true and $x_i = 0$ represents it being false. Each clause $c \in \mathcal{C}$ has an associated positive weight $w_c > 0$. Each literal in a clause is associated with a variable index and a sign $s \in \{+1, -1\}$, where $+1$ denotes a positive literal (x_i) and -1 denotes a negated literal ($\neg x_i$).

QUBO Derivation. Step 1 — Original Objective. The problem requires maximizing the total weight of satisfied clauses:

$$\max_{x \in \{0,1\}^n} \sum_{c \in \mathcal{C}} w_c \cdot \mathbb{I}(c \text{ is satisfied}), \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Since QUBO solvers minimize energy, we negate the objective:

$$\min_{x \in \{0,1\}^n} H_0(x) = -\sum_{c \in \mathcal{C}} w_c \cdot \mathbb{I}(c \text{ is satisfied}). \tag{2}$$

Step 2 — Constraints. The problem is unconstrained; any binary assignment is feasible. Thus, no constraint terms are introduced.

Step 3 — Algebraic Expansion of Satisfaction. A clause c is satisfied if at least one literal evaluates to true. Let L_c be the set of literals in clause c. For a literal $l \in L_c$ on variable i with sign s_l, its truth value is $y_l = \frac{1+s_l}{2} x_i + \frac{1-s_l}{2}(1 - x_i)$. The clause is unsatisfied if and only if $\prod_{l \in L_c}(1 - y_l) = 1$. Therefore,

$$\mathbb{I}(c \text{ is satisfied}) = 1 - \prod_{l \in L_c}(1 - y_l). \tag{3}$$

Substituting into H_0 yields:

$$H_0(x) = \sum_{c \in \mathcal{C}} w_c \prod_{l \in L_c} (1 - y_l) - \sum_{c \in \mathcal{C}} w_c. \tag{4}$$

We expand the product for clauses of size one and two. For a unit clause c with literal l on variable i:

$$1 - y_l = \frac{1 + s_l}{2} - s_l x_i. \tag{5}$$

For a binary clause c with literals l_1, l_2 on variables i, j with signs s_1, s_2:

$$(1 - y_{l_1})(1 - y_{l_2}) = \left(\frac{1 + s_1}{2} - s_1 x_i\right)\left(\frac{1 + s_2}{2} - s_2 x_j\right) = \frac{(1 + s_1)(1 + s_2)}{4} - \frac{s_2(1 + s_1)}{2} x_i - \frac{s_1(1 + s_2)}{2} x_j + \tag{6}$$

Step 4 — Combined Hamiltonian. Collecting all linear, quadratic, and constant terms across all clauses, the total energy function $H(x)$ takes the standard QUBO form:

$$H(x) = \sum_{i=1}^{n} Q_{i,i} x_i + \sum_{1 \le i < j \le n} Q_{i,j} x_i x_j + c. \tag{7}$$

Step 5 — Q Matrix and Offset. Reading off the coefficients from the expanded terms gives the general closed-form expressions for the QUBO parameters:

$$Q_{i,i} = \sum_{c \in \mathcal{C}_1(i)} -w_c s_{l_c} + \sum_{j \neq i} \sum_{c \in \mathcal{C}_2(i,j)} -\frac{s_{c,j}(1 + s_{c,i})}{2} w_c, \tag{8}$$

$$Q_{i,j} = \sum_{c \in \mathcal{C}_2(i,j)} w_c s_{c,i} s_{c,j} \quad \text{for } i < j, \tag{9}$$

$$c = -\sum_{c \in \mathcal{C}} w_c + \sum_{c \in \mathcal{C}_1} w_c \frac{1 + s_{l_c}}{2} + \sum_{c \in \mathcal{C}_2} w_c \frac{(1 + s_{c,i})(1 + s_{c,j})}{4}, \tag{10}$$

where $\mathcal{C}_1(i)$ denotes unit clauses containing variable i, $\mathcal{C}_2(i, j)$ denotes binary clauses containing both variables i and j, and $s_{c,i}, s_{c,j}$ are the signs of the literals on variables i and j in clause c.

Penalty Weight Justification. The Weighted Max-2-SAT problem is inherently unconstrained; every binary assignment corresponds to a valid truth valuation. Consequently, no penalty terms or penalty weights are required. The QUBO formulation directly encodes the objective function without auxiliary variables or constraint penalties.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^n$ recovers the optimal assignment because the energy function is constructed as a linear transformation of the clause-satisfaction weight. For any assignment x, the term $\sum_{c\in\mathcal{C}} w_c \prod_{l\in L_c}(1-y_l)$ evaluates exactly to the total weight of unsatisfied clauses. Thus, $H(x) = W_{\text{unsat}}(x) - W_{\text{total}}$, where $W_{\text{total}} = \sum_{c\in\mathcal{C}} w_c$ is constant. Minimizing $H(x)$ is therefore equivalent to minimizing the weight of unsatisfied clauses, which is mathematically identical to maximizing the weight of satisfied clauses. Since the domain is unconstrained, the global minimum of $H(x)$ must correspond to a feasible assignment that achieves the maximum possible satisfied weight.

Illustrative Example. Consider an instance with $n = 3$ variables and three clauses:

- $c_1 : (x_1 \vee \neg x_2)$ with weight $w_1 = 3$. Literals have signs $s_{1,1} = +1, s_{1,2} = -1$.
- $c_2 : (\neg x_1 \vee x_3)$ with weight $w_2 = 2$. Literals have signs $s_{2,1} = -1, s_{2,3} = +1$.
- $c_3 : (x_2)$ with weight $w_3 = 4$. Literal has sign $s_{3,2} = +1$.

Applying the general formulas:

$$Q_{1,1} = -\frac{(-1)(1+1)}{2}(3) - \frac{(1)(1-1)}{2}(2) = 3, \tag{11}$$

$$Q_{2,2} = -(4)(1) - \frac{(1)(1-1)}{2}(3) = -4, \tag{12}$$

$$Q_{3,3} = -\frac{(-1)(1+1)}{2}(2) = 2, \tag{13}$$

$$Q_{1,2} = (3)(+1)(-1) = -3, \quad Q_{1,3} = (2)(-1)(+1) = -2, \quad Q_{2,3} = 0. \tag{14}$$

The constant offset is:

$$c = -(3+2+4) + (4)\frac{1+1}{2} + (3)\frac{(2)(0)}{4} + (2)\frac{(0)(2)}{4} = -9 + 4 = -5. \tag{15}$$

The resulting QUBO energy function is:

$$H(x) = 3x_1 - 4x_2 + 2x_3 - 3x_1x_2 - 2x_1x_3 - 5. \tag{16}$$

Evaluating $H(x)$ over all $2^3 = 8$ assignments yields the minimum energy $H(1,0,1) = -5$. This corresponds to the assignment $x_1 = 1, x_2 = 0, x_3 = 1$, which satisfies c_1 and c_2 (total weight $3+2=5$) and leaves c_3 unsatisfied, confirming the formulation correctly recovers the optimal solution.